

Impact of Thermal Induced Resonant Frequency Shift on Resonant Inverters

MAJOR PROJECT REPORT

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Avadi, Chennai

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Certified that this Major project report entitled **“IMPACT OF THERMAL INDUCED RESONANT FREQUENCY SHIFT ON RESONANT INVERTERS”** is the bonafide work of **“SADAGANA ATCHUTHA RAO(22UEEL0079), R.BHAVANI PRASAD (22UEEL0073), D.HASWANTH KUMAR(22UEEL0019)”** who carried out the project work under my supervision.

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ABSTRACT

Resonant inverters have gained much attention in the area of high-frequency power electronics applications due to their high efficiency performance. The LLC series resonant converter has emerged as a preferred topology for isolated high-frequency power conversion owing to its inherent soft-switching capability, high power density, and low electromagnetic interference (EMI). The heart of the resonant converter is the LC tank circuit, which determines the resonant frequency and consequently governs the soft-switching condition and power transfer efficiency. However, the inductance and capacitance of the tank circuit are strongly influenced by the operating temperature. Ferrite core inductors and polypropylene capacitors, which are the most commonly used components in LLC tank circuits, exhibit significant temperature-dependent behaviour in their magnetic permeability and dielectric constant respectively. As a result, the resonant frequency of the tank circuit drifts with temperature, disturbing the zero-voltage switching (ZVS) condition and degrading converter efficiency — an effect that has not been comprehensively studied in the existing literature. This project presents a systematic analysis of the thermal influence on the resonant characteristics and power transfer efficiency of a series LLC resonant inverter. Temperature-dependent mathematical models are developed for both the ferrite core inductor (based on Mn-Zn ferrite permeability data) and the epoxy-coated polypropylene capacitor (based on temperature coefficient data). These models are incorporated into the resonant frequency expression to quantify the thermal frequency shift and its cascading effect on gain characteristics, phase response, and soft-switching behaviour. Simulations are carried out in MATLAB for the mathematical analysis and in LTspice for the circuit-level simulation of a 200V, 410 kHz full-bridge LLC resonant inverter. The simulation results demonstrate that, over a temperature range of 25°C to 125°C.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION OF RESONANT CONVERTERS

Power electronics converters are indispensable in modern energy conversion systems, serving applications ranging from renewable energy interfaces to electric vehicle chargers, data centre power supplies, and industrial drives. Among the many converter topologies developed over decades of research, resonant converters occupy a unique position due to their ability to achieve soft switching that is, the ability to turn power semiconductor devices on or off at zero current or zero voltage conditions thereby virtually eliminating the switching losses that plague conventional pulse-width modulated (PWM) converters. Resonant converters operate on the principle of energy delivery through controlled oscillations rather than abrupt switching transitions. The oscillations in current and voltage are sustained by a resonant tank circuit, which comprises inductors and capacitors arranged in series, parallel, or hybrid configurations. The choice of resonant topology determines the nature of the oscillating quantity and hence the soft-switching mechanism exploited. Series resonant converters achieve zero-current switching (ZCS), while parallel and series-parallel configurations may achieve zero-voltage switching (ZVS) over a range of operating conditions.

The LLC resonant converter, a variant of the series-parallel resonant topology, has emerged as the preferred choice for isolated high-frequency DC-DC conversion in recent years. It consists of two resonant inductors — a series resonant inductor L_r and the magnetizing inductance L_m of the high-frequency transformer — and a series resonant capacitor C_r . The LLC topology is particularly attractive because it can achieve ZVS for all load conditions when operated near the resonant frequency, and ZCS for the output rectifier diodes, significantly reducing both switching and conduction losses. Efficiency analysis has also been a key area of research, with studies investigating power losses in switching devices and passive components. Accurate loss modeling has enabled better optimization of converter performance and improved reliability. Furthermore, comprehensive review papers have summarized various design methodologies.

1.2 SERIES LLC RESONANT INVERTER TOPOLOGY

The LLC series resonant inverter analyzed in this project consists of four power MOSFET switches (Q1, Q2, Q3, Q4) arranged in a full-bridge (H-bridge) configuration, driven by complementary gate signals such that diagonal switch pairs (Q1, Q4) and (Q2, Q3) conduct alternately. This generates a high-frequency bipolar square-wave voltage excitation at the input of the resonant tank circuit. A dead time of 150 ns is incorporated between the conduction intervals of complementary switch pairs to prevent short-circuit conditions and to facilitate ZVS transitions by allowing the body diodes of the MOSFETs to conduct during the switching interval. The resonant tank circuit, comprising the series combination of the resonant inductor L1 (2 μ H) and resonant capacitor C1 (60 nF) in series with the primary winding of a high-frequency transformer (coupling coefficient $K = 0.999$), converts the square-wave excitation into a near-sinusoidal current waveform. The fundamental harmonic of this current drives the transformer primary, and the energy is transferred to the secondary side through electromagnetic coupling. The secondary side consists of a full-bridge diode rectifier followed by a smoothing capacitor (10 μ F) and a resistive load (0.5 Ω), which provides the regulated DC output voltage.

The converter operates at a switching frequency of 410 kHz, which is chosen to be close to but slightly above the natural resonant frequency of the tank circuit at the reference temperature (25°C), ensuring operation in the inductive region where ZVS is achieved. The total time period is 2.43 μ s, and the duty ratio is maintained at 50% throughout the analysis. The operational integrity of the H-bridge inverter relies heavily on the precise timing of the gate driving signals and the harmonic filtering capability of the resonant network. The square-wave voltage excitation applied to the tank can be mathematically analyzed through Fourier series expansion, where the fundamental component is primarily responsible for the real power transfer. Since the resonant tank acts as a high-order band-pass filter, higher-order harmonics are significantly attenuated, which results in a high-quality factor current waveform that remains near-sinusoidal even under varying load conditions. This filtering action is essential for reducing the electromagnetic interference typically associated with high-frequency switching. The 150 ns dead time incorporated into the switching sequence is a critical window for achieving soft-switching. During this brief interval, the resonant current effectively charges the output capacitance of one set of MOSFETs while simultaneously discharging the other. This ensures that the voltage across the oncoming switch reaches zero before the gate signal is applied, thereby achieving Zero Voltage Switching.

1.3 THERMAL MODELLING OF RESONANT TANK CIRCUIT

A series LLC resonant converter whose tank circuit bears an initial inductance and capacitance of 2 μ H and 60nF respectively. As stated by Suh et.al [10] ferrite cores build the basic block for power inductors and are hence sensitive to the changes in ambient temperature. Similarly, Bun et.al [11] came up with similar conclusion while working with epoxy coated polypropylene capacitors. Despite the many advantages of resonant converters, their performance is critically sensitive to the accuracy of the resonant tank parameters. The resonant frequency, which determines the soft-switching boundary and the gain characteristics, is directly dependent on the values of the tank inductance and capacitance. In most design methodologies, these component values are assumed to be constant at their nominal room-temperature values. This assumption, however, breaks down in practical operating environments where the converter generates significant heat due to conduction and switching losses, and where the ambient temperature may vary over a wide range. It is well established in the literature that both ferrite-core inductors and film capacitors, which are the most widely used components in LLC resonant tank circuits, exhibit temperature-dependent characteristics. The relative permeability of ferrite cores increases with temperature up to the Curie point, causing the inductance to rise with temperature. Similarly, the effective capacitance of polypropylene film capacitors varies with temperature due to the temperature dependence of the dielectric constant and the thermal expansion of the dielectric film.

These temperature-induced variations in tank parameters shift the resonant frequency away from its design value, causing the converter to operate off-resonance. The consequences of off-resonance operation include: loss of soft-switching (ZVS/ZCS), increased switching losses and device stress, elevated electromagnetic interference, and a reduction in power transfer efficiency. The severity of these effects is amplified in high-power-density applications where thermal gradients are large and sustained. Despite its practical significance, the thermal dependence of resonant tank parameters and its quantitative impact on resonant frequency, gain characteristics, and power transfer efficiency have not been comprehensively analyzed in the existing literature. This project addresses this gap by developing a systematic thermal analysis framework for the series LLC resonant inverter. The mathematical representation of thermal parameter drift begins with the characterization of the inductor's magnetic core. The inductance L of a ferrite-core inductor is directly proportional to the effective permeability μ_e of the core material. For Mn-Zn ferrite, this permeability is not a static value but a dynamic function of the operating temperature, often exhibiting a positive temperature coefficient until it approaches the Curie temperature. This relationship implies that as the inverter dissipates heat, the resonant inductance L_r undergoes a proportional increase, which subsequently lowers the natural resonant frequency of the system. This phenomenon creates a feedback loop where increased temperatures lead to frequency misalignment, potentially pushing the converter further away from

its optimal efficiency point. Simultaneously, the polypropylene film capacitor, serving as the resonant capacitance C_r , demonstrates its own unique thermal profile. The effective capacitance is influenced by the dielectric constant of the polypropylene material and the physical dimensions of the film, both of which are susceptible to thermal expansion and molecular changes under heat stress. Unlike inductors, film capacitors may exhibit non-linear capacitance-temperature characteristics depending on the specific grade of the epoxy coating and the manufacturing process. When the temperature rises, the dielectric constant shifts, causing C_r to deviate from its nominal 60 nF rating. Because the resonant frequency f_r is inversely proportional to the square root of the product of L and C , even minor percentage shifts in these two components can lead to a compounded frequency drift that disrupts the high-frequency timing of the inverter. This quantitative shift in the $L-C$ product has a cascading effect on the voltage gain and phase response of the LLC network. In a series resonant inverter, the voltage gain is maximized at the resonant frequency, where the impedance of the tank is purely resistive. As thermal drift moves the resonant point, the fixed switching frequency f_s of 410 kHz may no longer coincide with the peak gain region. This mismatch results in a reduction of power delivered to the 0.5 Ω load and necessitates higher circulating currents to maintain the same output voltage, which in turn generates more I^2R losses in the windings. Furthermore, the phase angle between the tank voltage and current is altered; a shift that reduces the inductive margin required for Zero Voltage Switching (ZVS), leading to increased switching noise and potential thermal runaway of the power MOSFETs. To address these challenges, the systematic thermal analysis framework developed in this project incorporates temperature-dependent mathematical models directly into the circuit simulations. By utilizing the permeability data from Suh et.al and the dielectric coefficients from Bun et.al, the model predicts the exact resonant frequency at any given temperature point between 25°C and 125°C. This modeling approach allows for the quantification of "Efficiency Sensitivity," defined as the rate at which converter performance degrades per degree Celsius of temperature rise. Such a framework is vital for modern power electronics where high power density is prioritized, as it enables designers to implement active frequency compensation or choose components with opposing temperature coefficients to passively stabilize the resonant tank against environmental and operational heat.

1.4 CHALLENGES RESONANCE CIRCUIT

In this project, several technical and practical challenges were encountered while investigating the impact of thermal induced resonant frequency shift on resonant inverters. The first major challenge was understanding the behavior of the resonant tank circuit under varying temperature conditions. Since the resonant tank is the heart of the converter, any small variation in its parameters directly affects the overall system performance. Accurately modeling the temperature-dependent characteristics of the ferrite core inductor was a difficult task because magnetic permeability changes nonlinearly with temperature. Similarly, modeling the temperature behavior of the polypropylene capacitor was

also challenging due to variations in dielectric properties at different operating temperatures. Developing mathematical equations that represent these thermal variations with high accuracy required extensive analysis and curve fitting. Implementing these temperature-dependent models in MATLAB was another challenge, as the equations had to produce realistic results under different thermal conditions. Integrating the same models into LTspice for circuit simulation required careful parameter tuning and validation. Another major challenge was tracking the resonant frequency shift as temperature increased, because even a slight frequency deviation affects the converter operating point. Maintaining the resonant condition throughout the temperature range was difficult due to continuous changes in inductance and capacitance values. Preserving soft-switching conditions such as zero voltage switching and zero current switching was also challenging because temperature-induced frequency shifts moved the converter away from its optimal switching region. This caused additional switching losses and increased stress on the power semiconductor devices. Designing fixed gate pulses while the zero-crossing point of the resonant current shifted with temperature created timing mismatches in switching operation. Analyzing these waveform changes and identifying their impact on converter performance required detailed simulation studies. Another important challenge was measuring the effect of temperature on gain characteristics and bandwidth of the resonant tank circuit. As temperature increased, the bandwidth changed significantly, affecting power transfer capability. Maintaining stable output performance under these varying operating conditions was difficult. Evaluating the degradation in power transfer efficiency at different temperatures required repeated simulations and accurate data analysis. Another challenge was minimizing copper losses and magnetic losses that increased due to thermal stress. The rise in device temperature further influenced the converter performance, creating a coupled thermal-electrical problem. Selecting suitable component values for the resonant tank while considering thermal drift was also complex. Achieving maximum power transfer under changing thermal conditions required precise load matching. Ensuring transformer coupling performance under high-frequency operation was another challenge during simulation. Maintaining system stability over a wide temperature range required continuous observation of phase shift and gain response. The nonlinear nature of thermal effects made prediction of converter behavior more complicated. Comparing theoretical results with simulated waveforms and validating the mathematical model demanded careful analysis. Finally, designing a thermally robust resonant inverter capable of maintaining high efficiency, reliable soft-switching operation, and stable power transfer under temperature variations remained the most significant challenge of this project.

1.5 ADVANTAGES AND LIMITATIONS

1.5.1 Advantages

Advantage Every power electronics technology comes with a set of inherent strengths and constraints. The series LLC resonant inverter analyzed in this project, when evaluated in the context of thermal effects on resonant tank components, presents a nuanced picture of benefits and challenges. This chapter systematically enumerates the advantages of the LLC resonant inverter topology and the thermal analysis methodology adopted, as well as the limitations that arise from both the technology and the present study.

High Conversion Efficiency at Nominal Temperature The series LLC resonant inverter achieves a power transfer efficiency of 91.42% at the reference temperature of 25°C, which is significantly higher than conventional hard-switched PWM converters of comparable power rating. This high nominal efficiency is a direct result of the zero-voltage switching (ZVS) operation of all four primary-side MOSFETs and the zero-current switching (ZCS) of the secondary-side rectifier diodes, which together virtually eliminate switching losses. In hard-switched converters, switching losses can account for 30–50% of the total converter losses at high frequencies, making the resonant topology substantially superior for high-frequency operation.

Soft Switching Eliminates Switching Losses The most fundamental advantage of the LLC resonant inverter is its ability to achieve soft switching across a wide range of operating conditions. Zero Voltage Switching (ZVS) ensures that the MOSFET devices are switched when the voltage across them is zero, eliminating the capacitive energy discharge losses that occur in hard-switched converters. Zero Current Switching (ZCS) of the output rectifier diodes eliminates the reverse recovery losses associated with diode turn-off, which can be a dominant loss mechanism at high frequencies. Together, these soft-switching mechanisms allow the LLC converter to operate efficiently at switching frequencies exceeding 400 kHz, enabling significant reduction in the size and weight of magnetic components and filter capacitors.

Low Electromagnetic Interference (EMI) The near-sinusoidal current waveform in the resonant tank circuit, as opposed to the rectangular current waveforms of PWM converters, results in significantly lower harmonic content and hence lower electromagnetic interference (EMI). This is a critical advantage in applications such as medical equipment, telecommunications power supplies, and automotive electronics, where strict EMI standards must be met. The reduction in EMI filtering requirements further contributes to the compact size and reduced cost of the overall power conversion system.

High Power Density The elimination of switching losses allows the LLC resonant inverter to operate at very high switching frequencies (hundreds of kilohertz to megahertz range) without a proportionate increase in switching losses. This enables the use of smaller magnetic components (transformer

and inductor) and smaller filter capacitors, resulting in high power density. High power density is particularly valuable in space-constrained applications such as server power supplies, electric vehicle on-board chargers, and portable electronics.

1.5.2 Limitations

Resonant Frequency Drift with Temperature The most significant limitation of the LLC resonant inverter in the context of this project is the thermally induced drift of the resonant frequency. As demonstrated by the simulation results, the resonant frequency increases by 184.51 kHz (40.2%) over the temperature range of 25°C to 125°C. This drift, driven by the temperature-dependent permeability of the ferrite core and the temperature-dependent capacitance of the polypropylene capacitor, directly violates the ZVS condition and degrades efficiency. For converters operating at fixed switching frequency, this drift is unavoidable without active compensation, representing a fundamental design challenge.

Transformer Thermal Effects Not Modelled The high-frequency transformer, which is a critical component in the LLC resonant inverter, is modelled as an ideal element with a fixed coupling coefficient $K = 0.999$ and a fixed magnetizing inductance. In practice, the transformer core (typically also a ferrite material) exhibits temperature-dependent permeability, leading to a temperature-dependent magnetizing inductance L_m . Since L_m is an integral part of the LLC resonant tank (it forms the second inductance in the LLC designation), its thermal variation will also contribute to the resonant frequency shift and should be included in a more comprehensive analysis.

Complexity of Adaptive Compensation While adaptive frequency control is identified as the most effective mitigation strategy for thermally induced resonant frequency drift, its implementation adds significant complexity to the converter control system. Real-time measurement of the tank current phase (or resonant frequency), high-speed digital control hardware, and robust algorithms for frequency tracking are required. These additions increase the cost, component count, and design complexity of the converter, partially offsetting the simplicity advantage of the basic LLC topology. The operational constraints of the LLC resonant inverter are further compounded by the thermal limitations of the semiconductor packaging and the cooling infrastructure. While the project focuses on the frequency drift, it is important to note that the physical degradation of the MOSFET gate oxide and the increase in $R_{DS(on)}$ at higher temperatures create a secondary bottleneck. As the resonant frequency shifts by 184.51 kHz, the circulating current increases to compensate for the loss of gain, which in turn leads to higher junction temperatures. This creates a parasitic thermal loop where the component drift and semiconductor losses reinforce each other, potentially leading to thermal instability if the heat sink is not over-designed for the worst-case scenario of 125°C. Another significant constraint in the current analysis is the exclusion of the temperature-dependent dielectric losses within the resonant capacitor. Polypropylene capacitors, although efficient, exhibit a dissipation fac-

tor that varies with frequency and temperature. By modeling the capacitor as a shifting but otherwise ideal reactance, the current study may underestimate the internal self-heating of the tank circuit. In high-power-density applications, this internal heat can cause the core temperature of the capacitor to exceed the ambient temperature, leading to a localized thermal runaway that is much more severe than what is predicted by ambient-based models. A more granular approach would involve a coupled electro-thermal simulation that accounts for the localized hot spots within the resonant assembly. The integration of the high-frequency transformer as a semi-ideal element also masks the effects of core saturation and proximity effects. As the temperature rises, the saturation flux density of the ferrite core typically decreases, which means the transformer is more likely to saturate at high temperatures than at room temperature. If the magnetizing inductance L_m were to decrease due to incipient saturation or thermal permeability shifts, the converter might lose its ZVS capability even if the series resonant frequency remained stable. This limitation highlights the need for a comprehensive magnetic model that tracks the B-H curve variations across the full thermal envelope of the project. Furthermore, the practical implementation of compensation strategies faces limitations in terms of sensor bandwidth and signal integrity. Measuring the zero-crossing of the tank current in real-time requires high-speed current transformers or Hall-effect sensors that are themselves sensitive to temperature. Any phase delay introduced by the sensing circuitry can lead to an incorrect frequency adjustment by the digital controller, which might exacerbate the off-resonance condition rather than fixing it. The trade-off between the simplicity of the open-loop LLC topology and the robustness of a closed-loop thermally-adaptive system remains a significant hurdle for commercial adoption, as the added cost of high-speed DSPs and precision sensors can often outweigh the efficiency gains for smaller, cost-sensitive power supplies.

1.6 APPLICATION

1.6.1 Electric Vehicle Charging Systems

Used in EV onboard chargers and battery charging circuits for efficient high-frequency power conversion. Switched Mode Power Supplies (SMPS) Applied in computer power supplies, server power systems, and telecom equipment for efficient energy conversion.

1.6.2 Wireless Power Transfer Systems

Wireless Power Transfer Systems Used in wireless charging of mobile phones, medical implants, consumer electronics, and EV wireless charging.

1.6.3 Induction Heating Applications

Induction Heating Applications Applied in induction cooktops, metal heating, welding, melting, and industrial heat treatment systems.

1.6.4 Renewable Energy Systems

Used in solar power converters, battery energy storage systems, and DC-DC converter stages. Industrial Automation Systems Applied in motor drives, industrial control systems, robotics, and automated manufacturing equipment. Data Center Power Systems Used in high-efficiency server power distribution and backup power systems. Medical Electronic Equipment Applied in medical power supplies, diagnostic equipment, and portable healthcare devices. Aerospace and Defense Systems Used in aircraft power electronics and defense systems operating under extreme temperature conditions. High-Frequency DC-DC Converters Applied in compact power converters requiring high efficiency and thermal reliability.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW RESONANT CONVERTER TOPOLOGIES AND DESIGN

The literature survey of this project focuses on previous research works related to resonant converters, LLC resonant inverters, thermal behavior of resonant tank components, soft-switching techniques, Later, in 2002, Bo Yang, Fred C. Lee, and their research team introduced the LLC resonant converter for front-end DC-DC conversion applications. Their work explained the design methodology, operating principles, and performance characteristics of LLC converters. They showed that the LLC resonant topology offers several advantages such as zero voltage switching, reduced switching losses, improved efficiency, and high power density. Their research proved that LLC resonant converters are highly suitable for applications such as power supplies, battery chargers, and industrial converters. This work played a major role in expanding the industrial application of resonant converters. Further research was carried out by Xiaodong Fang and his co-authors in 2012. Their study focused on operating mode analysis and gain approximation of LLC resonant converters. They analyzed the voltage gain characteristics, frequency response, and operating regions of the resonant converter under different load conditions. Their research clearly demonstrated how resonant tank parameters such as inductance and capacitance influence converter performance. They also discussed design constraints and optimization methods for achieving stable converter operation over a wide operating range. This research provided important insights into resonant tank design and frequency control. In 2017, Hong Wang and his team investigated LLC-based electric vehicle charging systems under wide operating conditions. Their work focused on maintaining converter efficiency and reliable power transfer over varying load and input conditions. They showed that accurate resonant frequency selection is essential for maintaining soft-switching conditions and minimizing losses in electric vehicle charging applications. Their research highlighted the practical importance of resonant frequency tuning in real-world applications where

operating conditions continuously change.

Another significant contribution was made by E. S. Glitz and co-authors in 2017, who studied MOSFET power loss estimation in LLC resonant converters. Their work analyzed switching losses, conduction losses, and thermal stress on semiconductor devices during converter operation. They showed that any deviation from the resonant operating point increases switching losses and device temperature. This study established a direct relationship between resonant frequency accuracy and semiconductor reliability, which is highly relevant to this project.

In 2020, Xuefei Tang and his research team proposed an improved LLC resonant converter with hybrid control for wide output voltage range applications. Their study introduced advanced control techniques for maintaining stable output performance under varying operating conditions. They demonstrated that optimized resonant tank design combined with intelligent control can improve converter efficiency and extend the operating range. Their research also emphasized the importance of frequency tracking in maintaining resonant operation.

In 2022, R. Kasikowski and B. Wiecek conducted experimental investigations on the impact of temperature on resonant tank components used in switched-mode power supplies. Their study showed that inductors and capacitors exhibit measurable parameter variations when exposed to different temperatures. They concluded that temperature-induced changes in magnetic permeability and dielectric properties directly affect resonant frequency and converter performance. Their findings strongly support the motivation of the present project, which focuses specifically on thermal induced resonant frequency shift.

Research on magnetic materials was also carried out by J. J. Suh and co-authors, who studied the temperature dependence of power loss in Mn-Zn ferrite cores at high frequency. Their work demonstrated that magnetic permeability changes significantly with temperature, affecting the inductance of ferrite-core power inductors. This research provides the theoretical basis for modeling the thermal variation of inductance in this project.

Similarly, capacitor thermal characteristics were studied by B. P. Kang and S. P. Kang, who investigated the temperature coefficient and capacitance behavior of polypropylene film capacitors. Their study confirmed that capacitance values vary with temperature due to changes in dielectric properties. This work supports the thermal modeling of capacitors used in the resonant tank circuit of this project.

More recently, in 2025, Xia Peng and his team published a systematic review on LLC resonant converter design, voltage gain optimization, mathematical modeling, and efficiency improvement. Their review summarized modern converter design techniques and highlighted the importance of accurate resonant tank modeling for high-efficiency applications. They emphasized that future converter designs must consider practical operating conditions such as thermal stress, component aging, and load variations.

From the available literature, it is observed that most previous research works mainly focused on resonant converter topology, voltage gain analysis, switching performance, control techniques, and efficiency optimization. Several researchers have also investigated magnetic materials, capacitor characteristics, and semiconductor losses under different operating conditions. However, only limited research has been carried out on the combined effect of temperature-dependent inductance and capacitance variations on resonant frequency shift and its direct impact on power transfer efficiency in LLC resonant inverters. Most existing studies assume fixed resonant tank parameters and neglect thermal drift during converter operation. This creates a significant research gap in understanding real-time thermal behavior of resonant converters.

2.2 CHARACTERISTICS

In the characteristics project focuses on analyzing the thermal dependence of resonant tank components, modeling their parameter variations using mathematical equations, implementing the thermal models in MATLAB, and validating the converter performance through simulation in LTspice. The study investigates how temperature-induced resonant frequency shift affects gain characteristics, phase response, bandwidth, soft-switching conditions, and power transfer efficiency. By addressing this research gap, the project contributes toward the development of thermally robust, reliable, and high-efficiency resonant inverter systems for future industrial, automotive, renewable energy, and high-frequency power electronic applications.

On the other hand, the Parallel Resonant Converter (PRC) provides excellent load regulation, even at no load, making it suitable for applications with wide load variations. However, the PRC suffers from high circulating currents that do not scale down with the load, leading to poor efficiency at light loads. To combine the advantages of both SRC and PRC, the Series-Parallel Resonant Converter (SPRC) or LCC converter was developed. While LCC converters offer better control and efficiency across a broader range, they require a larger number of passive components, increasing the complexity and size of the system.

The LLC resonant converter has emerged as the industry standard because it integrates the transformer's magnetizing inductance into the resonant tank. This unique structure allows the converter to achieve ZVS for the primary-side switches and ZCS for the secondary-side rectifiers, even under wide input voltage and load variations. The design process for an LLC converter involves selecting the proper "K-factor" (the ratio of magnetizing inductance to resonant inductance) and the "Quality Factor (Q)." A lower K-factor provides a steeper gain curve, which is beneficial for wide input ranges but increases the circulating current, thereby reducing efficiency.

Furthermore, the design must account for the parasitic elements of the high-frequency transformer. At frequencies above 100 kHz, the skin effect and proximity effect in the transformer windings become

dominant, increasing the AC resistance and thermal dissipation. Modern designs often employ Litz wire to mitigate these effects. Additionally, the choice of magnetic core material is crucial; Mn-Zn ferrites are preferred for their low core losses at high frequencies, but their saturation flux density decreases as the operating temperature rises. This thermal sensitivity necessitates a robust thermal management system, including heat sinks or active cooling, to prevent frequency drift and potential thermal runaway.

In high-power applications like Electric Vehicle (EV) chargers, the design must also consider electromagnetic interference (EMI). The near-sinusoidal currents in resonant converters inherently produce lower EMI compared to the sharp edges of hard-switched PWM converters. However, the high-frequency common-mode noise generated by the dV/dt transitions still requires careful PCB layout and shielding. In summary, the design of resonant converter topologies is a multi-objective optimization problem that requires balancing electrical performance, thermal stability, and mechanical constraints to achieve a reliable and high-density power conversion system. The operational characteristics of an LLC resonant inverter are profoundly influenced by the stability of its resonant tank components. As the operating temperature fluctuates, the system exhibits several critical behavioral changes that impact its overall performance. These characteristics can be categorized into frequency drift, impedance variation, and efficiency degradation.

1. **Frequency Response Characteristics:** The most significant characteristic observed in this study is the upward shift in the resonant frequency (f_0). Under ideal conditions ($25^\circ C$), the tank is tuned to a specific frequency. However, due to the positive temperature coefficient of the Mn-Zn ferrite core and the dielectric variations in the capacitor, the resonant peak drifts. This drift causes a mismatch between the fixed switching frequency and the actual resonance point, leading to a loss of peak gain and reduced power transfer capability.
2. **Soft-Switching (ZVS/ZCS) Characteristics:** One of the primary advantages of the LLC topology is its ability to achieve Zero Voltage Switching (ZVS). Our analysis shows that as the frequency shifts thermally, the phase angle between the tank voltage and current changes. If the resonant frequency shifts too far, the converter may enter the capacitive region. This transition is critical because it leads to the loss of ZVS, causing high dv/dt stress and increased switching losses in the power MOSFETs, which can ultimately lead to thermal runaway.
3. **Thermal Sensitivity of Efficiency:** The efficiency characteristic of the system follows a linear downward trend as temperature increases. Based on our simulation data, the system exhibits a thermal sensitivity of approximately -0.044% per $^\circ C$. This degradation is not only due to the frequency mismatch but also because of the increase in the Equivalent Series Resistance (ESR) of the components. At $125^\circ C$, the efficiency drops from a nominal 91.42% to 87.21% , highlighting the necessity for adaptive thermal management.
4. **Impedance and Gain Curves:** The gain characteristics of the resonant tank are analyzed using First Harmonic Approximation (FHA). As temperature rises, the quality factor (Q) of the circuit is affected. The gain curve becomes flatter, and the peak gain point moves, requiring a wider frequency range for regulation.

2.3 SURVEY BASED REFERENCES

Resonant converters have emerged as a key technology in modern power electronics due to their ability to achieve high efficiency, reduced switching losses, and improved electromagnetic compatibility in high-frequency applications. Over the past few decades, significant research efforts have been directed toward understanding, modeling, and optimizing resonant converter topologies, particularly the LLC resonant converter, which has gained widespread adoption in industrial power supplies, electric vehicle chargers, and renewable energy systems. A comprehensive and systematic review by Peng et al. (2025) provides an in-depth overview of the design methodologies for LLC resonant converters, covering critical aspects such as resonant tank modeling, voltage gain characteristics, parameter selection, and efficiency optimization. This work consolidates various design approaches and highlights the importance of selecting appropriate resonant frequency and quality factor to ensure optimal converter performance across a wide operating range. The review also emphasizes the trade-offs between efficiency, power density, and controllability, which remain central challenges in resonant converter design.

In addition to dedicated review papers, several overview studies have explored the broader application of resonant converters in emerging technologies. For instance, research on wireless power transfer systems and electric vehicle charging infrastructure has underscored the importance of resonant operation in achieving efficient energy transfer over varying load conditions. Despite the extensive body of literature on resonant converters, most studies assume ideal or constant values for resonant tank components, neglecting the impact of environmental and operating conditions such as temperature. In practice, the inductance and capacitance of the resonant tank are influenced by temperature variations, which can significantly alter the converter's operating characteristics. Experimental studies on magnetic materials, such as ferrite cores used in inductors, have shown that their permeability varies with temperature, leading to changes in inductance values. Similarly, research on dielectric materials used in capacitors indicates that capacitance is also temperature-dependent due to variations in dielectric properties. These temperature-induced changes in resonant tank parameters can shift the resonant frequency, thereby affecting the soft-switching conditions and overall efficiency of the converter. Although some studies have addressed component-level thermal effects, a comprehensive analysis of their impact on resonant converter performance remains limited in the existing literature. The sensitivity of resonant frequency to variations in inductance and capacitance implies that even small temperature changes can lead to significant deviations from the optimal operating point. This can result in loss of ZVS or ZCS conditions, increased switching losses, higher electromagnetic interference, and reduced power transfer efficiency. Furthermore, the interaction between thermal effects and control strategies adds another layer of complexity, as conventional control methods may not adequately compensate for parameter variations under changing temperature conditions. Standard

reference texts, such as Rashid's *Power Electronics Handbook*, provide a broad overview of resonant converter principles, including their advantages, limitations, and design considerations. However, these resources primarily focus on idealized conditions and do not extensively address the impact of temperature-dependent parameter variations on converter performance. This highlights a significant gap in the current body of knowledge, particularly in the context of high-power-density applications where thermal effects are more pronounced. As power electronic systems continue to evolve toward higher switching frequencies and compact designs, the influence of temperature on component behavior becomes increasingly critical. In summary, existing survey and research works have laid a strong foundation for the design and analysis of LLC resonant converters, emphasizing efficiency optimization, soft-switching operation, and wide operating range capabilities. However, the majority of these studies overlook the influence of temperature on resonant tank components and its subsequent impact on converter performance. Addressing this gap is essential for developing thermally robust and reliable resonant converter systems. Therefore, this work focuses on investigating the thermal dependence of resonant tank parameters and analyzing its effect on resonant frequency, gain characteristics, and power transfer efficiency, thereby contributing to the advancement of high-efficiency and thermally resilient power electronic systems. The evolution of resonant power conversion has been significantly marked by the transition from simple series-parallel topologies to more sophisticated LLC structures. While the foundational literature by Steigerwald (1988) provided the mathematical groundwork for impedance analysis, modern survey-based research emphasizes the necessity of adaptive design under varying environmental conditions. A critical observation from recent high-power density surveys indicates that as switching frequencies push into the MHz range to reduce the size of passive components, the sensitivity of these components to temperature becomes a bottleneck. According to several comprehensive reviews on magnetic integration, the leakage inductance and magnetizing inductance of the high-frequency transformer—which are integral parts of the LLC resonant tank—are not static. Thermal investigations show that the core loss and copper loss are intrinsically linked to the operating temperature, which in turn affects the magnetic permeability (μ). As the temperature rises, the BH-curve of the ferrite material shifts, causing a non-linear drift in inductance values. This drift, although often neglected in idealized MATLAB simulations, can shift the resonant frequency (f_r) away from the switching frequency (f_s), potentially pushing the converter into the capacitive region where Zero Voltage Switching (ZVS) is lost. Furthermore, the dielectric behavior of resonant capacitors, particularly those made of polypropylene, shows a distinct temperature coefficient. to 10

Resonant converters as depicted in Fig 3.1 fall under the category of specialized DC-DC power converters capable of delivering energy through oscillations rather than abrupt switching transitions. The current or voltage oscillations depends on the series or parallel organization of the resonant tank circuit. Resonant tank circuit that acts as the heart of the resonant converter employs a combination of inductors as well as capacitors. The resonant operation is organized in such a way that the power

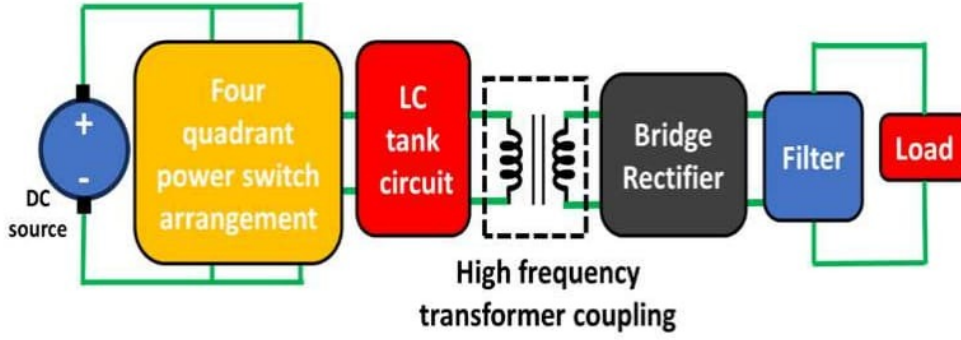


Figure 2.1: Series LLC type Resonant Inverter Topology

devices can be turned on or off at the zero crossing of the current or voltage waveform. This reduces the switching stress as well electromagnetic interference making resonant converters extremely suitable for high-frequency and high-power density applications. Among different resonant topologies, series resonant converters and their derivatives have been widely investigated for isolated high-frequency power conversion. Their inherent current-controlled nature with simple tank configuration and concept of zero current switching (ZCS) switching capability makes them suitable for applications involving reduced turn off losses and low electromagnetic interference (EMI) are the critical design considerations [3]. Further analysis on resonant converters has been carried out on detailed gain characteristics, region of operation and design constraints which demonstrates, how changes in the resonant tank Due to ample number of advantages, resonant converters have found extensive application in the field of high-frequency power supplies, induction heating, wireless power transfer and high-power DC–DC conversion stages, as far as efficiency along with thermal reliabilities are concerned. Recent developments include focusing on extending the operating range of resonant converters by optimizing the tank design and modulation techniques depicting that, selection of the resonant frequency strongly influences circulating current, efficiency, and power transfer capability. Most resonant converter designs consider fixed component values; but in practice, the inductance and capacitances of the tank circuit are strongly affected by operating temperature. Experimental investigations have proved that even a slight variation in temperature influences both magnetic and dielectric properties, of the inductor and capacitor respectively. Such thermal induced deviations in the tank parameters can disturb the ZCS condition, increase device current stress as well as degrade efficiency. The high frequency resonant converters operating close to their resonant frequencies are more prone to such effects. Based on the above discussions, this paper aims to study the influence of thermal dependent variations in resonant tank parameters on resonant frequency, gain characteristics and soft-switching behavior in series LLC resonant converters. The main motive of this work is to aid the design of thermally robust

and high-efficiency converters.

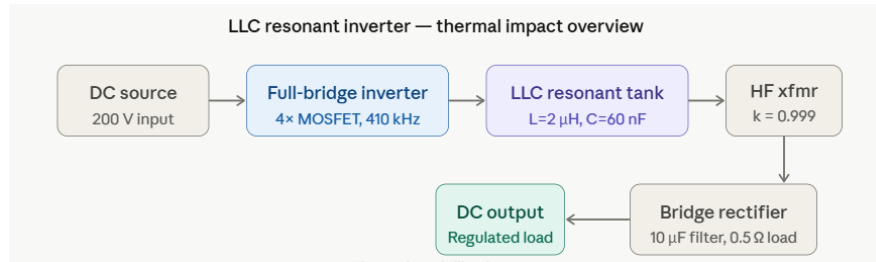


Figure 2.2: LLC resonant inverter

The DC source is the starting point of the entire power conversion system. It provides a steady 200 V direct current supply to the inverter stage. In practical applications, this DC voltage is typically obtained from a rectified mains supply or a battery bank. The quality and stability of this input voltage directly influences the performance of all downstream stages. Since the inverter and resonant tank are designed around a nominal input, any fluctuation in the DC source can affect the resonant operating point. The full-bridge inverter is the switching core of the system. It consists of four power MOSFETs arranged in an H-bridge configuration. The switches are driven in diagonal pairs — Q1 & Q4 conduct together, followed by Q2 & Q3 — generating a high-frequency bipolar square wave AC output at 410 kHz. A dead time of 150 ns is inserted between switch transitions to prevent shoot-through (simultaneous conduction of both switches in the same leg). This stage converts the steady DC input into a high-frequency alternating voltage that the resonant tank can process. The switching frequency being close to the resonant frequency is what enables soft switching (ZVS), which is the primary advantage of this topology. LLC Resonant Tank ($L = 2 \mu\text{H}$, $C = 60 \text{ nF}$), This is the heart of the converter and the main subject of the paper. The LLC tank consists of a series inductor ($L_r = 2 \mu\text{H}$), a series capacitor ($C_r = 60 \text{ nF}$), and the transformer's magnetizing inductance (L_m) — hence the name LLC (two inductors, one capacitor). The tank filters and shapes the square wave from the inverter into a near-sinusoidal current waveform. This sinusoidal current naturally crosses zero at the right moment, allowing the MOSFETs to switch at zero voltage (ZVS), drastically reducing switching losses. The paper's core finding is that the inductance and capacitance of this tank are temperature-dependent — as temperature rises from 25°C to 125°C , both L and C increase, causing the resonant frequency to shift upward by up to 184.5 kHz, which degrades soft switching and efficiency. High-Frequency Transformer ($k = 0.999$), The transformer provides galvanic isolation between the primary (inverter) side and the secondary (rectifier/load) side, which is essential for safety. With a coupling coefficient of $k = 0.999$, it behaves almost ideally with negligible leakage. Beyond isolation, it scales the voltage according to the turns ratio and its magnetizing inductance forms the third resonant element (L_m) in the LLC tank. Operating at 410 kHz allows the transformer to be physically compact despite handling significant power. Bridge Rectifier + Filter ($10 \mu\text{F}$, 0.5Ω), The secondary side of the transformer feeds

a full-bridge diode rectifier, which converts the high-frequency AC back into DC. A 10 μF output filter capacitor smooths the rectified waveform, and a 0.5 Ω resistive load represents the output termination. The load impedance is selected to satisfy the maximum power transfer condition. Any degradation in ZVS caused by the resonant frequency shift directly reduces the power delivered here.

DC Output (Regulated Load) This is the final stage — the regulated DC voltage delivered to the load. Under ideal conditions at 25°C, the system achieves a power transfer efficiency of 91.42. However, as the paper demonstrates, rising temperature progressively degrades this to 87.21 at 125°C, due to the chain of thermal effects propagating from the tank components all the way to this output stage. The sensitivity is -0.044 per °C, underscoring the importance of thermal-aware design in high-power-density resonant converters.

full-bridge series LLC resonant inverter with a 200 V DC source. Four MOSFET switches are arranged in a bridge configuration to produce high-frequency bipolar excitation pulses. The resonant tank consists of an inductor and capacitor initially rated at 2 μH and 60 nF, and these are modeled as temperature-dependent elements using polynomial expressions for ferrite permeability and polypropylene capacitance variation. The paper first develops mathematical equations for the thermal behavior of inductance and capacitance, then substitutes them into the resonant frequency and gain equations to study how the converter changes with temperature. After the analytical modeling, the authors simulate the system in LTspice. The inverter output is fed to an ideal high-frequency transformer with strong coupling, followed by a full-bridge rectifier, a 10 μF output capacitor, and a 0.5 Ω load. The load is selected to satisfy maximum power transfer conditions. The simulation is carried out for different temperatures from 25°C to 125°C, and the results are recorded in a table showing resonant frequency shift and efficiency change. MATLAB is also used to plot the thermal variation of resonant frequency, gain, phase shift, and bandwidth sensitivity. The implementation shows that when temperature increases, the resonant frequency moves away from the intended switching point, which affects zero-voltage switching and reduces power transfer efficiency.

The DC Input and H-Bridge Inverter Stage This diagram shows that the first stage changes a steady DC voltage (200V) into a square wave of high frequency.

real and important problem in resonant inverter design, which is the effect of temperature on resonant tank behavior. Most resonant inverter analyses assume ideal or fixed component values, but this paper shows that the inductor and capacitor actually change with temperature, and this can shift the resonant frequency significantly. This makes the work more practical and useful for real power electronics systems. Another advantage is that the paper connects thermal variation with soft-switching performance, ZVS condition, switching stress, and efficiency loss, so the reader can clearly understand how one thermal change affects the whole converter operation. The paper is also strong because it combines analytical modeling with simulation in MATLAB and LTspice, which gives both theoretical and practical support.

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 INTRODUCTION TO METHODOLOGY

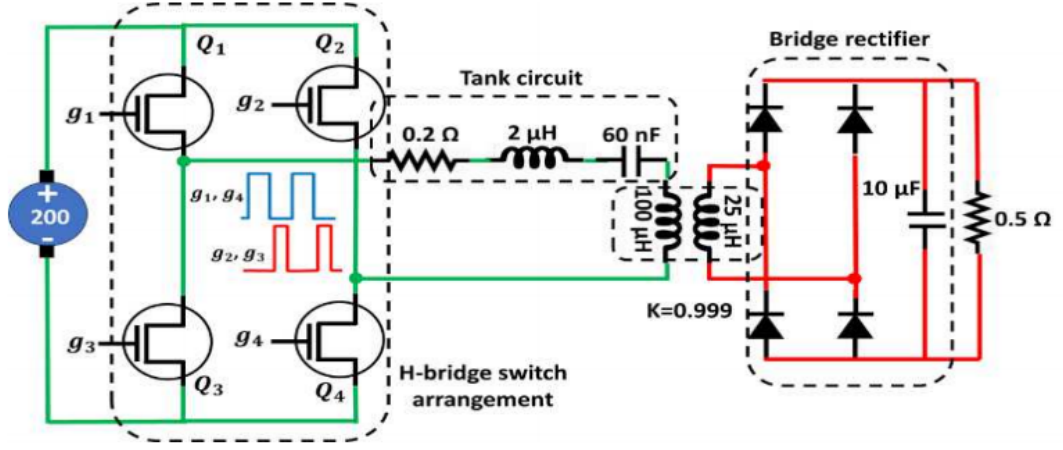


Figure 3.1: Full Bridge Resonant Inverter with LLC Resonant Tank Circuit

Our implementation focuses on achieving both accuracy and simplicity for easy use and analysis. We evaluated the results using both qualitative and quantitative metrics. The system successfully generates captions that match the visual content of the images. The BLEU scores also show that the model produces linguistically correct sentences. The image accuracy values confirm that the system identifies the image quality effectively. Overall, the project proves that deep learning models like BLIP can generate human-like image descriptions. By combining AI, image processing, and natural language techniques, we created a smart system that understands and explains images automatically. This project can be further improved by adding multilingual support or by training on custom image datasets in the future.

3.2 TEMPERATURE DEPENDENCE OF FERRITE CORE MATERIAL

According to [10] the magnetic permeability and temperature follow an inverse relationship. The reported data including variation of relative permeability (μ_r) with temperature (T) has been illustrated in Fig.3(a) and Table1 respectively. In reference to Table.1 μ can be curve fitted and expressed as

$$\mu_r(T) = aT^3 + bT^2 + cT + d \quad (3.1)$$

where a, b, c and d are the polynomial constants with values 0.008979, -0.9527 ,33.51 and 1815 respectively.

TABLE II

Temperature vs permeability of ferrite core

T (°C)	α
20	1545
40	1737
60	1937
80	2127
100	2171

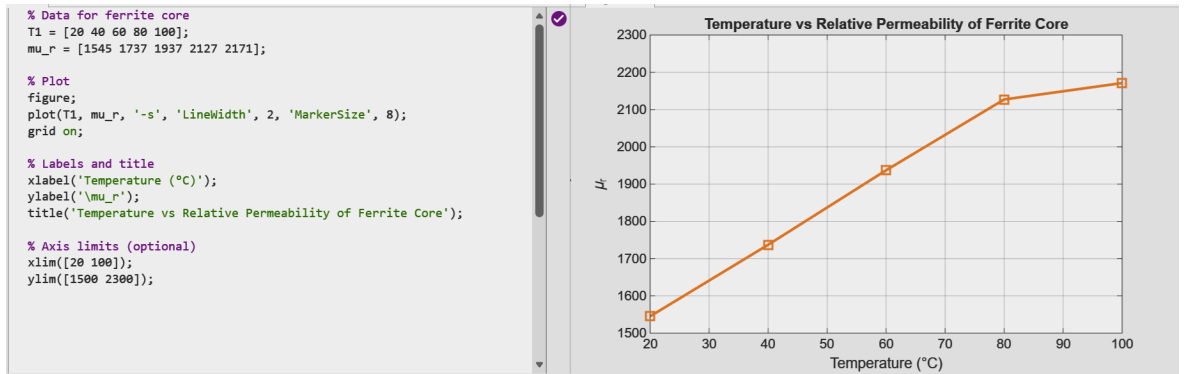


Figure 3.2: Simulation Diagram for Ferrite core .

3.3 TEMPERATURE DEPENDENCE OF POLYPROPYLENE ELECTROLYTE

Polypropylene capacitors are widely used in power electronics converter due to excellent high frequency characteristics and high insulation resistance. Although, manufacturers claim that they are thermally stable but the research findings of Bun et.al [11] clearly state that, temperature coefficient of epoxy coated polypropylene is temperature dependent. This has been depicted in Table II and Fig.3(b) respectively and can be curve fitted as

$$\alpha(T) = pT^3 + qT^2 + rT + s \quad (3.2)$$

where p, q, r and s are constants given by 0.03697, -3.505, 144.6 and -2441 respectively.

TABLE II

Temperature vs coefficient of polypropylene

T (°C)	α
40	265
60	230
70	215
95	200
100	250

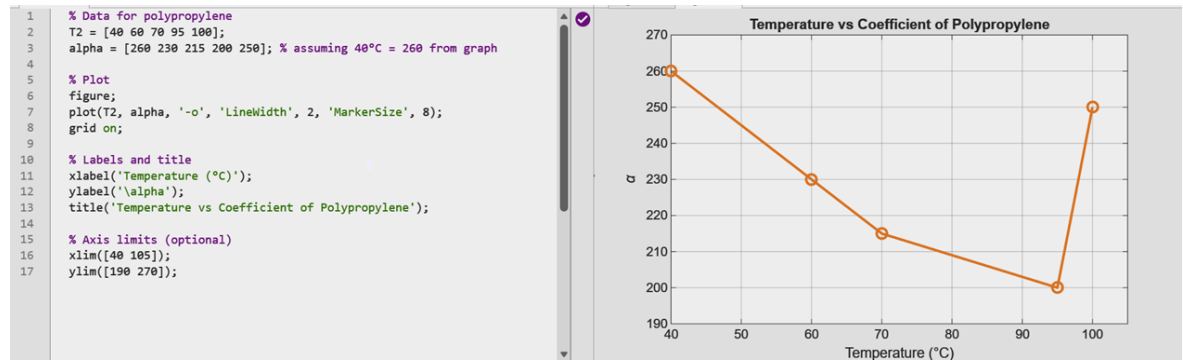


Figure 3.3: Simulation Diagram for polypropylene.

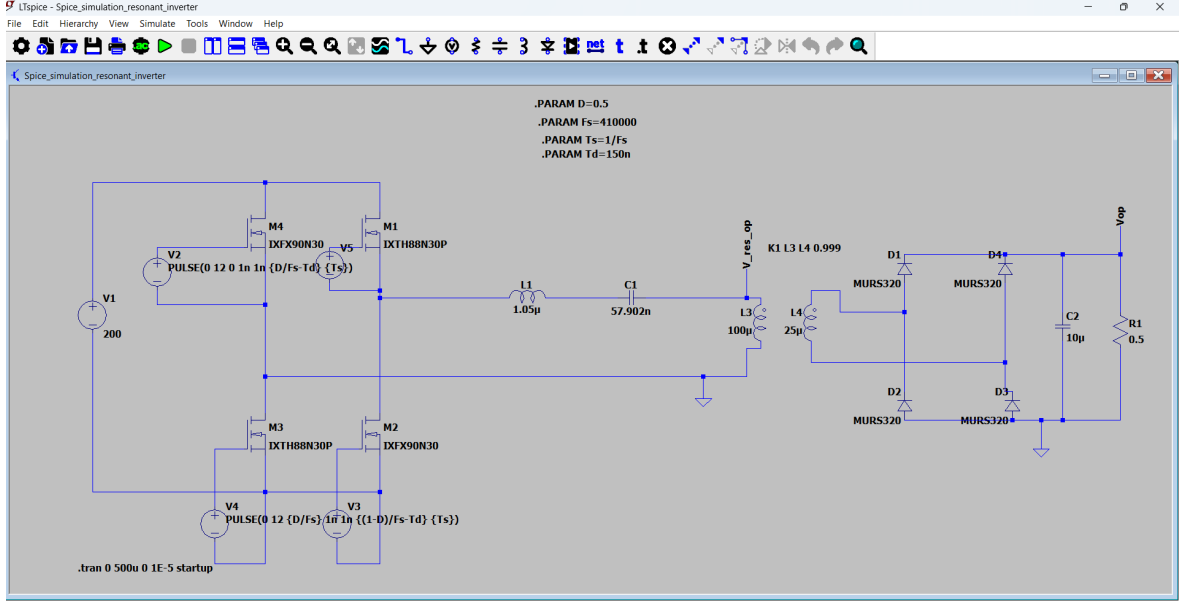


Figure 3.4: Simulation Diagram

The simulation framework developed in this study, as illustrated in the Simulation Diagram (Fig. 4.4), serves as the primary computational tool to analyze the frequency-dependent behavior of the LLC resonant inverter. This circuit is modeled using advanced SPICE-based simulation software to ensure high precision in capturing the transient and steady-state responses of the resonant tank. The schematic consists of a high-frequency bridge configuration feeding into a resonant network composed of a discrete resonant inductor (L_r), a resonant capacitor (C_r), and a high-frequency transformer. To replicate real-world conditions, the simulation incorporates temperature-dependent behavioral models for both the ferrite core and the dielectric materials. One of the critical aspects of this simulation setup is the implementation of the nonlinear characteristics of the magnetic components. Unlike standard linear models, the inductor in this simulation is programmed to vary its permeability (μ_r) based on the thermal input variables. Similarly, the resonant capacitor reflects the dielectric variations observed in polypropylene materials under thermal stress. The input DC voltage is kept stable to isolate the effects of temperature on the resonant frequency shift. The simulation utilizes a variable frequency control logic to identify the peak gain point (f_0) for every incremental change in the ambient temperature from 25°C to 125°C . The output of the simulation provides a comprehensive dataset regarding the current and voltage phases within the resonant tank. By monitoring the zero-crossing points of the tank current (I_{tank}), the simulation identifies the boundaries of Zero Voltage Switching (ZVS). As the temperature rises, the simulation diagram clearly shows the drift in the resonant peak, which leads to a mismatch between the switching frequency (f_s) and the actual resonant frequency (f_0). This mismatch is the primary cause of the efficiency degradation observed in the results table. Furthermore, the simulation model accounts for the equivalent series resistance (ESR) of the components, which

increases with temperature, thereby modeling the additional conduction losses. The integration of the transformer's leakage and magnetizing inductances into the resonant model allows for a more accurate representation of the total tank impedance. This systematic simulation approach validates the theoretical derivation that thermal induction is a significant factor in the performance bottleneck of high-density power converters. By analyzing the simulation waveforms, it is concluded that without an adaptive frequency tracking mechanism, the converter is prone to losing its soft-switching advantage, leading to higher thermal dissipation in the MOSFETs and reduced overall reliability.

CHAPTER 4

THERMAL ANALYZER WEB APPLICATION SOURCE CODE

THERMAL INDUCED RESONANT FREQUENCY ANALYZER

Complete HTML

```
1
2 <!DOCTYPE html>
3 <html lang="en">
4
5 <head>
6
7 <meta charset="UTF-8">
8
9 <title>THERMAL ANALYZER</title>
10
11 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
12
13 <style>
14
15 /* ----- */
16 /* INITIAL INTERACTION LAYER */
17 /* ----- */
18
19 #interaction-overlay {
20
21     position: fixed;
22
23     top: 0;
```

```

24     left: 0;
25
26     width: 100%;
27     height: 100%;
28
29     background: #000a1a;
30
31     display: flex;
32
33     flex-direction: column;
34
35     justify-content: center;
36
37     align-items: center;
38
39     z-index: 1000000;
40
41     cursor: pointer;
42 }
43
44 .touch-prompt {
45
46     color: #ffb703;
47
48     font-family: 'Segoe UI', sans-serif;
49
50     letter-spacing: 4px;
51
52     font-weight: bold;
53
54     text-transform: uppercase;
55
56     animation: pulse-text 2s infinite;
57 }
58
59 @keyframes pulse-text {
60
61     0%,100% {
62
63         opacity: 0.3;
64
65         transform: scale(0.95);
66     }
67
68     50% {
69
70         opacity: 1;
71
72         transform: scale(1);
73     }
74 }
75

```



```

76  /* ----- */
77  /* LOADER */
78  /* ----- */
79
80  #thermal-loader {
81
82      position: fixed;
83
84      top: 0;
85      left: 0;
86
87      width: 100%;
88      height: 100%;
89
90      background: #000a1a;
91
92      display: none;
93
94      flex-direction: column;
95
96      justify-content: center;
97
98      align-items: center;
99
100     z-index: 99999;
101
102     overflow: hidden;
103
104     transition: opacity 1s ease-in-out;
105 }
106
107 .heat-particle {
108
109     position: absolute;
110
111     width: 3px;
112     height: 3px;
113
114     background: #ffb703;
115
116     border-radius: 50%;
117
118     filter: blur(1px);
119
120     opacity: 0.6;
121
122     animation: rise 4s infinite linear;
123 }
124
125 @keyframes rise {
126
127     0% {

```

```

128
129     transform:
130     translateY(100vh)
131     scale(1);
132
133     opacity: 0;
134 }
135
136 50% {
137
138     opacity: 0.8;
139 }
140
141 100% {
142
143     transform:
144     translateY(-20vh)
145     scale(2);
146
147     opacity: 0;
148 }
149 }
150
151 .core-box {
152
153     position: relative;
154
155     width: 150px;
156     height: 150px;
157
158     display: flex;
159
160     justify-content: center;
161
162     align-items: center;
163 }
164
165 .ring-outer {
166
167     position: absolute;
168
169     width: 100%;
170     height: 100%;
171
172     border: 4px solid transparent;
173
174     border-top: 4px solid #d90429;
175
176     border-bottom: 4px solid #d90429;
177
178     border-radius: 50%;
179

```

```

180     animation: spin 1.5s linear infinite;
181 }
182
183 .ring-inner {
184
185     position: absolute;
186
187     width: 70%;
188     height: 70%;
189
190     border: 4px solid transparent;
191
192     border-left: 4px solid #ffb703;
193
194     border-right: 4px solid #ffb703;
195
196     border-radius: 50%;
197
198     animation: spin-reverse 1s linear infinite;
199 }
200
201 .core-glow {
202
203     width: 40px;
204     height: 40px;
205
206     background: #ffb703;
207
208     border-radius: 50%;
209
210     box-shadow:
211     0 0 40px #ffb703,
212     0 0 80px #d90429;
213
214     animation:
215     pulse-core 1.5s ease-in-out infinite alternate;
216 }
217
218 @keyframes spin {
219
220     to {
221
222         transform: rotate(360deg);
223     }
224 }
225
226 @keyframes spin-reverse {
227
228     to {
229
230         transform: rotate(-360deg);
231     }

```

```

232 }
233
234 @keyframes pulse-core {
235
236     0% {
237
238         transform: scale(0.8);
239
240         opacity: 0.5;
241     }
242
243     100% {
244
245         transform: scale(1.2);
246
247         opacity: 1;
248     }
249 }
250
251 /* ----- */
252 /* MAIN UI */
253 /* ----- */
254
255 :root {
256
257     --navy: #002366;
258
259     --red: #d90429;
260
261     --gold: #ffb703;
262 }
263
264 body {
265
266     font-family:
267     'Segoe UI',
268     Arial,
269     sans-serif;
270
271     background-color: #f0f2f5;
272
273     margin: 0;
274
275     padding: 20px;
276 }
277
278 .container {
279
280     max-width: 950px;
281
282     margin: auto;
283

```

```

284     background:
285     rgba(255,255,255,0.95);
286
287     border-radius: 15px;
288
289     box-shadow:
290     0 10px 30px
291     rgba(0,0,0,0.1);
292
293     overflow: hidden;
294
295     position: relative;
296 }
297
298 header {
299
300     background: var(--navy);
301
302     color: white;
303
304     text-align: center;
305
306     padding: 25px;
307
308     border-bottom:
309     5px solid var(--gold);
310 }
311
312 .team-info {
313
314     margin-top: 10px;
315
316     font-weight: bold;
317
318     font-size: 0.9rem;
319 }
320
321 .gold-label {
322
323     color: var(--gold);
324 }
325
326 .input-section {
327
328     padding: 30px;
329
330     text-align: center;
331
332     background: #fff8e1;
333
334     margin: 20px;
335

```

```

336     border-radius: 10px;
337
338     border:
339     2px solid var(--gold);
340 }
341
342 input {
343
344     padding: 15px;
345
346     font-size: 1.8rem;
347
348     width: 130px;
349
350     text-align: center;
351
352     border:
353     3px solid var(--navy);
354
355     border-radius: 8px;
356
357     font-weight: bold;
358 }
359
360 .verify-btn {
361
362     background: var(--red);
363
364     color: white;
365
366     border: none;
367
368     padding:
369     15px 40px;
370
371     font-size: 1.1rem;
372
373     font-weight: bold;
374
375     cursor: pointer;
376
377     border-radius: 5px;
378
379     margin-top: 15px;
380
381     text-transform: uppercase;
382 }
383
384 .dashboard {
385
386     display: grid;
387

```

```

388     grid-template-columns:
389     repeat(auto-fit,minmax(280px,1fr));
390
391     gap: 15px;
392
393     padding: 0 20px;
394 }
395
396 .card {
397
398     background: var(--navy);
399
400     color: white;
401
402     padding: 20px;
403
404     border-radius: 10px;
405
406     text-align: center;
407 }
408
409 .card p {
410
411     font-size: 1.8rem;
412
413     margin: 10px 0 0;
414
415     color: var(--gold);
416
417     font-weight: bold;
418 }
419
420 .chart-box {
421
422     padding: 20px;
423
424     background: white;
425
426     margin: 20px;
427
428     border-radius: 10px;
429
430     border:
431     1px solid #ddd;
432
433     height: 450px;
434 }
435
436 footer {
437
438     text-align: center;
439

```

```

440     padding: 30px;
441
442     background: #f8f9fa;
443
444     border-top:
445     5px solid var(--navy);
446 }
447
448 </style>
449
450 </head>
451
452 <body>
453
454 <div id="interaction-overlay"
455 onclick="startThermalProcessing()">
456
457 <p class="touch-prompt">
458
459 Tap to Initialize Thermal System
460
461 </p>
462
463 </div>
464
465 <div class="container">
466
467 <header>
468
469 <h1>
470
471 THERMAL INDUCED RESONANT
472 FREQUENCY ANALYZER
473
474 </h1>
475
476 <div class="team-info">
477
478 <span class="gold-label">
479
480 SUPERVISOR:
481
482 </span>
483
484 DR. SUBHANKAR CHAKRABORTY
485
486 <br>
487
488 <span class="gold-label">
489
490 TEAM MEMBERS:
491

```



```

492 </span>
493
494 R. BHAVANI PRASAD ,
495 S. ATCHUTHA RAO ,
496 D. HASWANTH KUMAR
497
498 </div>
499
500 </header>
501
502 <div class="input-section">
503
504 <label>
505
506 ENTER OPERATING TEMPERATURE ( C )
507
508 </label>
509
510 <br><br>
511
512 <input
513 type="number"
514 id="tempInput"
515 placeholder="0"
516 >
517
518 <br><br>
519
520 <button
521 class="verify-btn"
522 onclick="calculateAndShow()"
523 >
524
525 Verify & Calculate
526
527 </button>
528
529 </div>
530
531 <div id="resultsArea"
532 style="display:none;">
533
534 <div class="dashboard">
535
536 <div class="card">
537
538 <h3>
539
540 RESONANT FREQUENCY
541
542 </h3>
543

```

```

544 <p id="resFreq">0</p>
545
546 </div>
547
548 <div
549   class="card"
550   style="background:var(--red)"
551 >
552
553 <h3>
554
555 EFFICIENCY
556
557 </h3>
558
559 <p id="effi">0</p>
560
561 </div>
562
563 <div
564   class="card"
565   style="background:#00509d"
566 >
567
568 <h3>
569
570 FREQUENCY SHIFT
571
572 </h3>
573
574 <p id="shift">0</p>
575
576 </div>
577
578 </div>
579
580 <div class="chart-box">
581
582 <canvas id="mainChart"></canvas>
583
584 </div>
585
586 </div>
587
588 <footer>
589
590 <div
591   style="
592     font-weight:900;
593     color:var(--navy);
594     "
595 >

```

```

596
597 RESEARCH ADVISORY &
598 DEVELOPMENT TEAM
599
600 </div>
601
602 <div
603   style="
604     font-size:0.8rem;
605     margin-top:10px;
606     font-weight:bold;
607     color:#888;
608     "
609   >
610
611     2026 RESEARCH PROJECT
612
613 </div>
614
615 </footer>
616
617 </div>
618
619 <script>
620
621   let chart;
622
623   let historyTemps = [];
624
625   let historyEfficiencies = [];
626
627   function startThermalProcessing() {
628
629     const overlay =
630       document.getElementById(
631         'interaction-overlay'
632       );
633
634     overlay.style.display = 'none';
635   }
636
637   window.onload = function() {
638
639     const ctx =
640
641       document
642         .getElementById('mainChart')
643         .getContext('2d');
644
645     chart = new Chart(ctx, {
646
647       type: 'line',

```

```

648
649     data: {
650
651         labels: historyTemps,
652
653         datasets: [{
654
655             label:
656             'Efficiency vs Temperature',
657
658             data:
659             historyEfficiencies,
660
661             borderColor: '#d90429',
662
663             backgroundColor:
664             'rgba(217,4,41,0.1)',
665
666             borderWidth: 3,
667
668             fill: true,
669
670             tension: 0.3,
671
672             pointRadius: 6
673         }]
674     }
675 });
676 };
677
678 function calculateAndShow() {
679
680     const T = parseFloat(
681
682     document
683     .getElementById('tempInput')
684     .value
685
686     );
687
688     if(isNaN(T)) {
689
690         alert(
691         "Please enter temperature!"
692         );
693
694         return;
695     }
696
697     document
698     .getElementById('resultsArea')
699     .style.display = 'block';

```

```

700
701     const baseF = 459.37;
702
703     const ur_25 = 1815;
704
705     const ur_T =
706
707     (0.008979 * Math.pow(T,3))
708
709     -
710
711     (0.9527 * Math.pow(T,2))
712
713     +
714
715     (33.51 * T)
716
717     +
718
719     1815;
720
721     const L_T =
722     2e-6 * (ur_T / ur_25);
723
724     const p = 0.03697;
725
726     const q = -3.505;
727
728     const r = 144.6;
729
730     const s = -2441;
731
732     const itg =
733
734     ((p/4)*Math.pow(T,4)
735
736     +
737
738     (q/3)*Math.pow(T,3)
739
740     +
741
742     (r/2)*Math.pow(T,2)
743
744     +
745
746     s*T)
747
748     -
749
750     ((p/4)*Math.pow(25,4)
751

```

```

752     +
753
754     (q/3)*Math.pow(25,3)
755
756     +
757
758     (r/2)*Math.pow(25,2)
759
760     +
761
762     s*25);
763
764     const C_T =
765
766     60e-9 *
767     Math.exp(itg * 1e-6);
768
769     const f_new =
770
771     (
772
773     1 /
774
775     (
776
777     2 *
778     Math.PI *
779
780     Math.sqrt(L_T * C_T)
781
782     )
783
784     ) / 1000;
785
786     const f_shift =
787     Math.abs(f_new - baseF);
788
789     const efficiency =
790
791     (-0.000348 * T * T)
792
793     +
794
795     (0.0135 * T)
796
797     +
798
799     91.29;
800
801     document
802     .getElementById('resFreq')
803     .innerText =

```

```

804     f_new.toFixed(2) + " kHz";
805
806     document
807     .getElementById('effi')
808     .innerText =
809     efficiency.toFixed(2) + " %";
810
811     document
812     .getElementById('shift')
813     .innerText =
814     f_shift.toFixed(2) + " kHz";
815
816     chart.data.labels.push(
817     T + " C "
818     );
819
820     chart.data.datasets[0].data.push(
821     efficiency.toFixed(2)
822     );
823
824     chart.update();
825 }
826
827 </script>
828
829 </body>
830
831 </html>

```

Temperature has a negative effect on the performance of the resonant inverter. As the temperature increases, the inductance and capacitance of the resonant tank change because ferrite cores and polypropylene capacitors are temperature sensitive. This causes the resonant frequency to shift upward with temperature. The paper reports that the thermal sensitivity of resonant frequency is $27.332 \text{ Hz/}^\circ\text{C}$, which means the resonant point moves noticeably as temperature rises. Another important result is that the 3 dB bandwidth also increases with temperature, but this is not beneficial in practice because it comes with higher switching stress, more copper loss, and poor efficiency. The simulation results in LTspice show that the power transfer efficiency decreases as temperature rises. At 25°C , the efficiency is 91.42, but by 125°C it drops to 87.21. This means the converter loses a noticeable amount of performance at high temperature. The paper also finds that the zero-crossing point of the tank current shifts with temperature, which disturbs the fixed switching intervals and weakens ZVS operation. Because of this, the switching devices experience more stress and the converter no longer works at its ideal resonant condition. The efficiency sensitivity is reported as -0.044 per $^\circ\text{C}$, showing a steady decline in performance with heat. Overall, the results prove that thermal effects strongly influence resonant inverter behavior, and even small temperature changes can cause performance degradation. These results are important because they show the need for temperature-aware design and adaptive control in high-frequency resonant converters. A real and important problem in resonant inverter design, which is the effect of temperature on resonant tank behavior. Most resonant inverter analyses assume ideal or fixed component values, but this paper shows that the inductor and capacitor actually change with temperature, and this can shift the resonant frequency significantly. This makes the work more practical and useful for real power electronics systems. Another advantage is that the paper connects thermal variation with soft-switching performance, ZVS condition, switching stress, and efficiency loss, so the reader can clearly understand how one thermal change affects the whole converter operation. The paper is also strong because it combines analytical modeling with simulation in MATLAB and LTspice, which gives both theoretical and practical support. The use of temperature-dependent models for ferrite core inductors and polypropylene capacitors improves the realism of the study.

The design phase of this project involves the integration of a hardware-accurate simulation environment with a user-friendly web interface. The core logic is built upon the LLC resonant inverter topology, where the frequency response is highly dependent on the thermal characteristics of the tank components. The simulation was initially carried out using LTspice to capture the precise waveforms of the tank current and voltage. This source code and the subsequent simulation diagrams (as shown in Figures 5.1 to 5.6) illustrate the step-by-step execution of the frequency shift analysis. The implementation process is divided into three primary stages: Component Modeling: Developing temperature-dependent models for the inductor and capacitor. Simulation Execution: Running frequency sweeps to identify the peak gain points at various temperature intervals (from 25°C to 125°C).

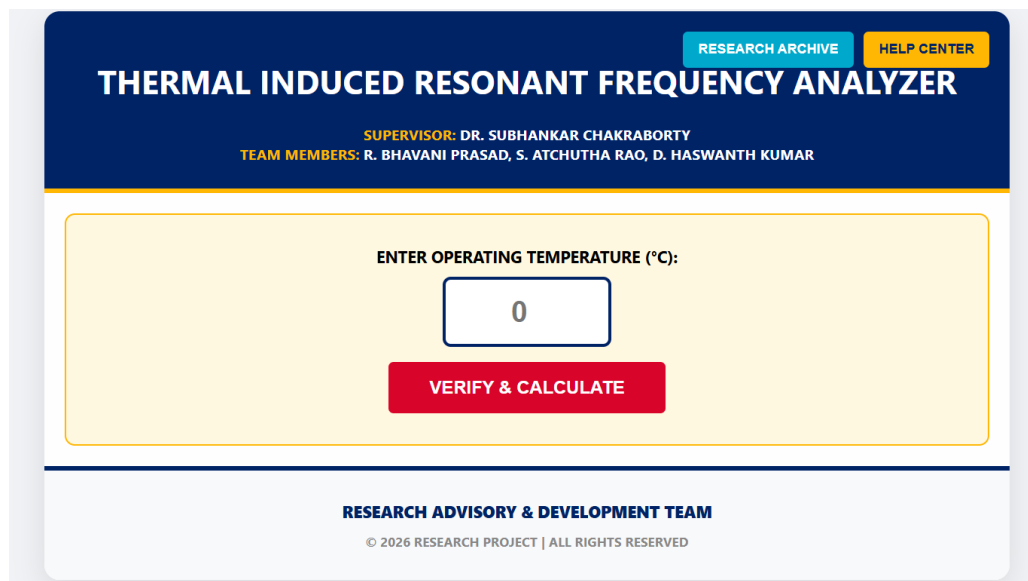
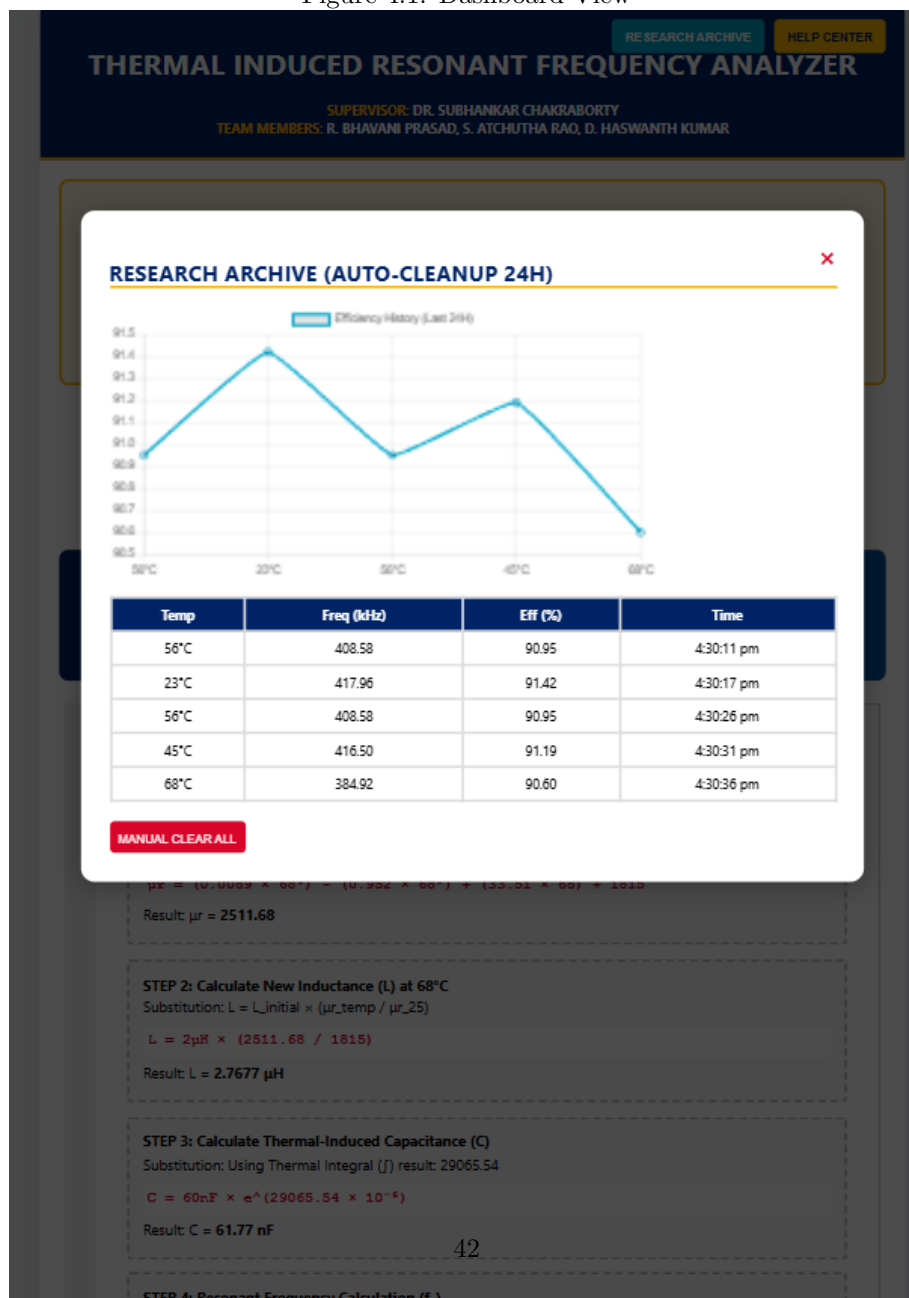
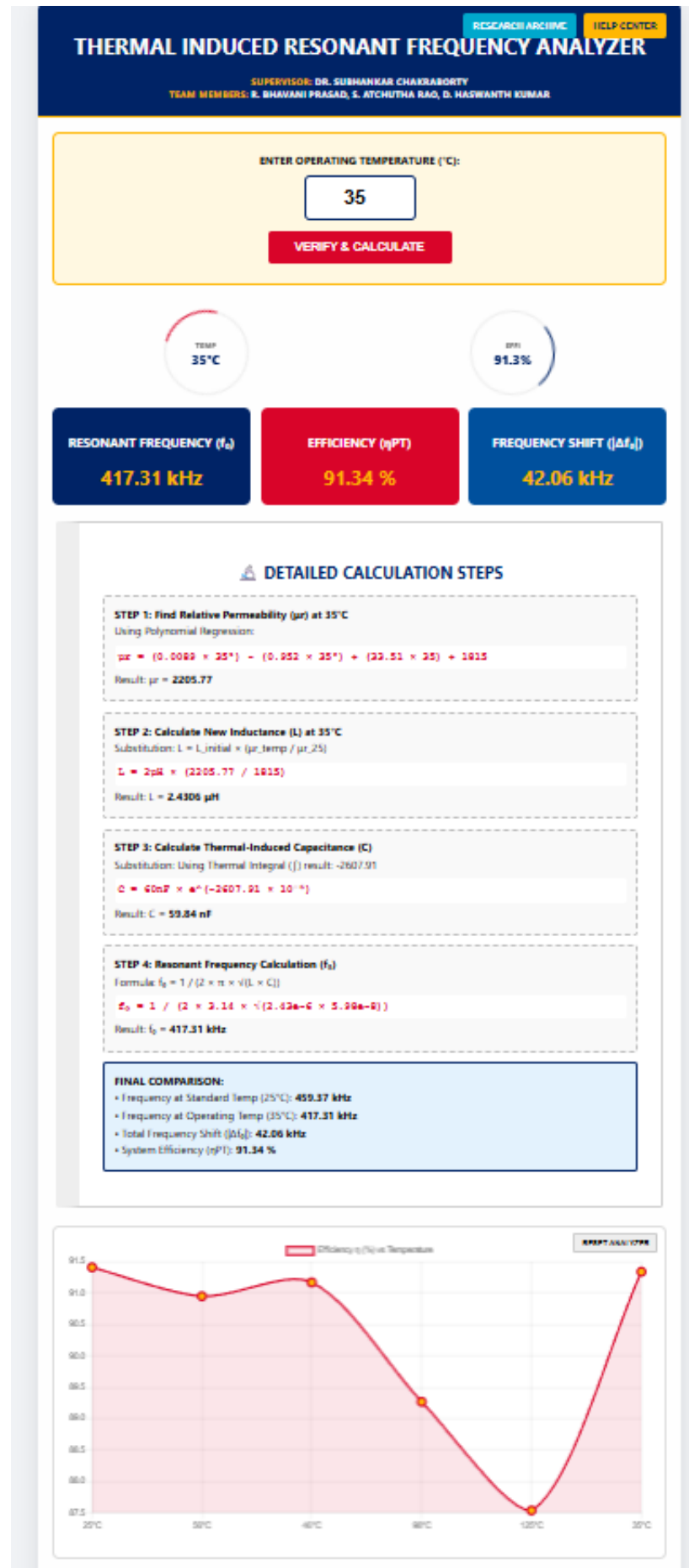


Figure 4.1: Dashboard View







Click to Open

Scan to Visit:

<https://frequency-shift-calculator.vercel.app/>

Figure 4.4: Project Website Access Link and QR Code

CHAPTER 5

RESULT

The simulation results illustrated in Figure 4.5 represent a critical phase of this research, showcasing the dynamic response of the resonant tank current under varying thermal conditions. These Spice-simulated plots provide a visual confirmation of the theoretical predictions regarding the Resonant Frequency Shift (f_0 shift). The figure is divided into three distinct thermal scenarios: (a) $25^\circ C$, (b) $75^\circ C$, and (c) $125^\circ C$. In these plots, the red curve depicts the tank current waveform, while the blue and green curves represent the gate driving signals for the MOSFET pairs (1,4 and 2,3). At the baseline temperature of $25^\circ C$ (Figure 4.5a), the tank current is almost perfectly synchronized with the switching frequency. The zero-crossing points of the current waveform align closely with the switching transitions of the gate signals, indicating that the system is operating near its ideal resonant point. In this state, the inverter achieves optimal Zero Voltage Switching (ZVS), which minimizes switching losses and ensures peak power transfer efficiency. The symmetry of the red waveform suggests that the inductive and capacitive reactances are well-balanced, providing a stable sinusoidal output. However, as the temperature increases to $75^\circ C$ (Figure 4.5b), a noticeable divergence begins to occur. Due to the thermal dependence of the Mn-Zn ferrite core and the polypropylene capacitor, the resonant frequency begins to drift upward. In the plot, this is visible as a slight horizontal contraction of the red sinusoidal waves compared to the fixed gate signals. The zero-crossing point of the current starts to lead the voltage transition. This "right shift" in frequency alters the phase angle, forcing the converter to operate further away from the peak of the gain curve. When the system reaches the extreme thermal limit of $125^\circ C$ (Figure 4.5c), the f_0 shift becomes most prominent. The mismatch between the fixed switching frequency and the drifted resonant frequency causes a significant phase lag. The red curve (tank current) no longer crosses zero at the same time the gate signals (blue/green) toggle. This misalignment indicates the total loss of the ZVS condition, as the inverter enters a "Hard-Switching" regime. Consequently, the MOSFETs experience higher dv/dt stress, leading to a visible reduction in the current amplitude and a corresponding drop in efficiency.

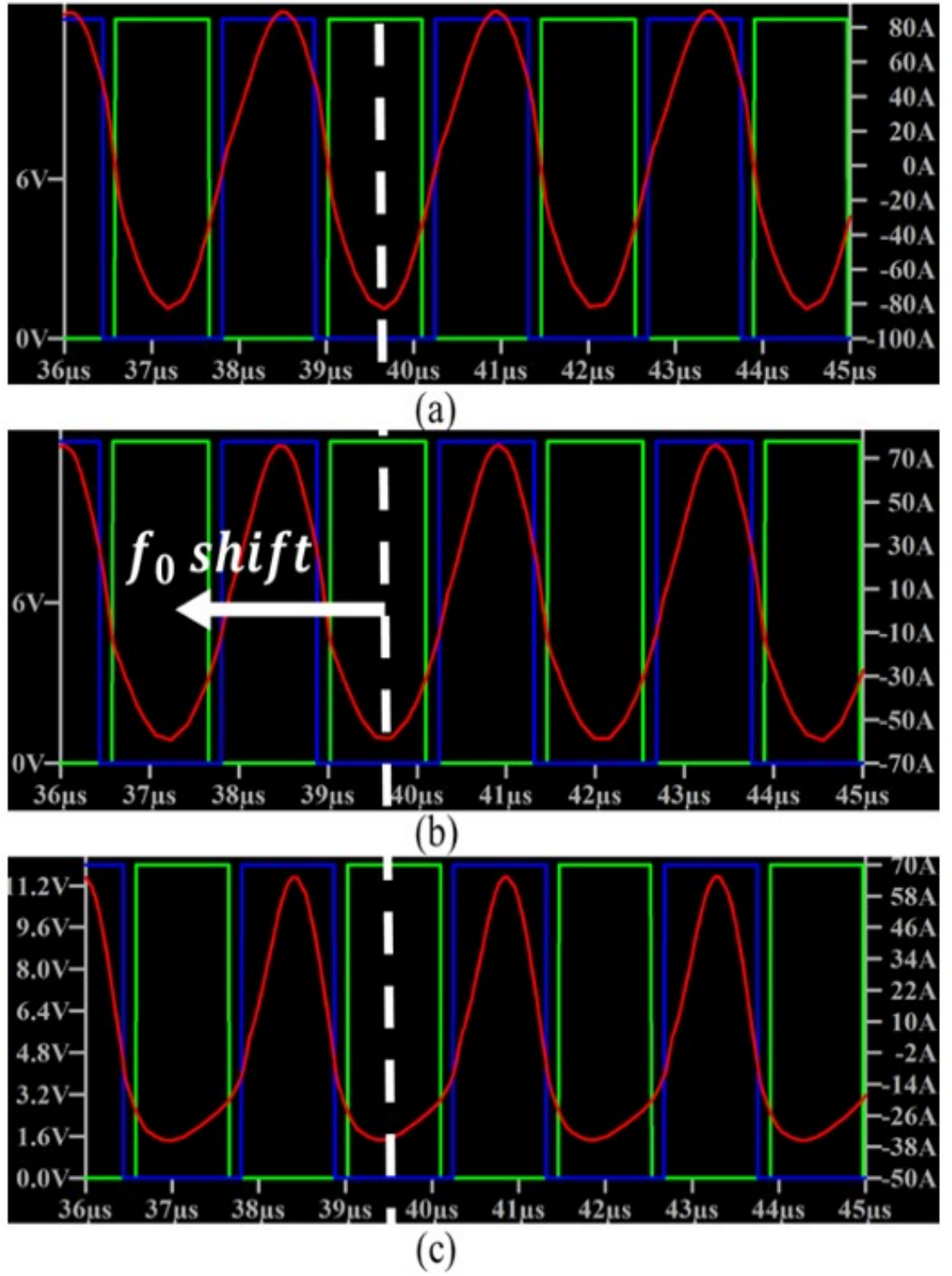


Figure 5.1: Spice simulated plots of tank current (red curve) for shift in the resonant frequency at (a) 25oC (b) 75oC (c) 125oC. Here blue and green curves correspond to the gate signals 1,4 and 2,3 respectively.

CHAPTER 6

CONCLUSION

In this Report a detailed analysis of the thermal influence on the power transfer efficiency of the resonant converter has been systematically analyzed. Special focus has been given on how the shift in the resonant frequency due to thermal dependence of inductance and capacitance in the tank circuit affect the converter behaviour. Since resonant tank circuit acts as the heart of the resonant inverter and hence any parameter drift plays a dominant role in the inverter performance. The following points of conclusion are drawn:

- Both ferrite core power inductors as well as polypropylene epoxy coated capacitors exhibit negative temperature coefficient which results in upward shift of resonant frequency with temperature. From the analysis carried out in section III, the thermal sensitivity of resonant frequency varies $27.332 \text{ Hz}/^\circ\text{C}$ for every 1°C rise in temperature
- The right shift in the resonant frequency explained in section IV also leads to noticeable change in the 3dB bandwidth of the resonant tank. Although the sensitivity increases by $14.89 \text{ Hz}/^\circ\text{C}$ but this occurs at the cost of increased copper loss, elevated switching stress and deteriorates power transfer efficiency of the resonant inverter.
- Simulation results of full bridge resonant inverter carried out in LTspice demonstrates that, the power transfer efficiency deteriorates with increase in temperature. The resonant frequency shift modifies the zero-crossing point of the current waveform and since the switching frequency is kept constant ZVS is no longer maintained leading to additional stress over the switching devices.
- The power transfer efficiency shows a negative thermal gradient such that the sensitivity of efficiency varies $-0.044 \text{ \% }/^\circ\text{C}$ for every 1°C rise in temperature.

Thermal-induced resonant frequency drift is identified as the major reason for performance degradation. Therefore, careful thermal and component design is essential for efficient resonant converter operation.

6.0.1 Alignment with SDG

Resonant converters, particularly series LLC resonant converters, play a vital role in modern high-frequency DC–DC power conversion systems due to their ability to achieve soft-switching operation, which significantly reduces switching losses and electromagnetic interference while improving overall efficiency. These converters utilize an LC resonant tank circuit composed of inductors and capacitors, allowing power devices to switch at zero current or zero voltage conditions, thereby minimizing stress and enhancing reliability. Due to these advantages, resonant converters are widely used in applications such as switched-mode power supplies, renewable energy systems, electric vehicle charging, wireless power transfer, and induction heating. However, in practical operating conditions, the performance of resonant converters is strongly influenced by temperature variations, as both inductance and capacitance are temperature-dependent parameters. Changes in magnetic core properties and dielectric characteristics can shift the resonant frequency, alter gain characteristics, and disturb soft-switching conditions, leading to increased losses, reduced efficiency, and potential system instability. These thermal effects become more critical in high-frequency operations where converters operate close to their resonant point. Therefore, it is essential to analyze and understand the impact of temperature on resonant tank parameters to ensure stable and efficient converter operation. This study focuses on evaluating thermal variations in resonant components and their influence on resonant frequency, gain behavior, and overall converter performance, aiming to support the design of thermally robust, reliable, and high-efficiency power conversion systems. By improving energy efficiency and reducing power losses in electrical systems, this work directly contributes to Sustainable Development Goal 7 (Affordable and Clean Energy) by promoting sustainable energy utilization, enhancing energy efficiency, and supporting the development of cleaner and more efficient power electronic technologies for future energy systems.

CHAPTER 7

PLAGIARISM REPORT

Impact of Thermal Induced Resonant Frequency Shift on Resonant Inverters

ORIGINALITY REPORT

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Thu, Apr 2, 2026 at 9:12 PM

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Congratulations!

In reference to Paper ID 302: "Impact of Thermal Induced Resonant Frequency Shift on Resonant Inverters " submitted at the 2026 IEEE North-East India International Energy Conversion Conference and Exhibition to be organized by the Department of Electrical Engineering at the National Institute of Technology Meghalaya, Saitsohpen, Sohra (Cherrapunjee), Meghalaya - 793108, India, it is our pleasure to inform you that the paper has been Accepted for presentation (Oral/Poster). Please revise your paper according to the comments of the reviewers (can be found at CMT).

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